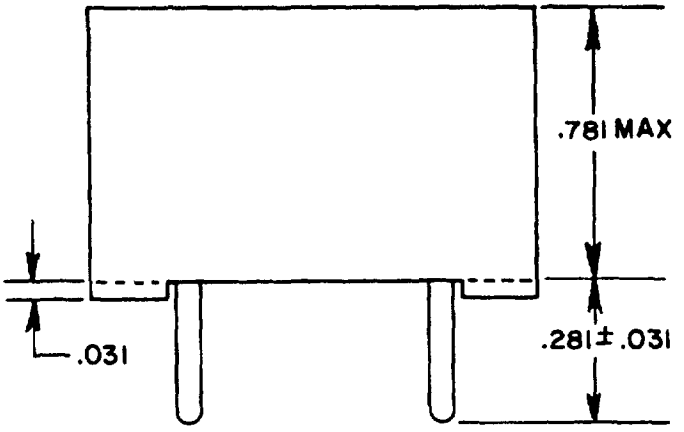
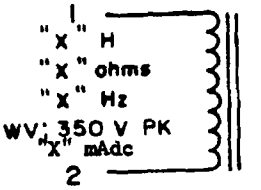
MILITARY SPECIFICATION SHEET

TRANSFORMERS AND INDUCTORS
(AUDIO, POWER AND HIGH-POWER PULSE),
INDUCTORS, HIGH Q, TF5R20ZZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

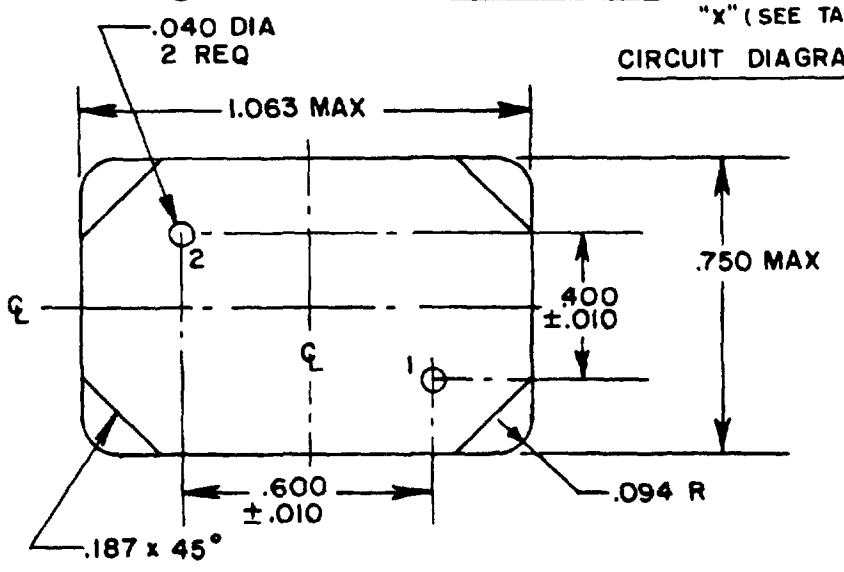
The complete requirements for procuring the inductors described herein shall consist of this document and the latest issue of Specification MIL-T-27.





MAX ALTITUDE 75,000 FT
"X" (SEE TABLE I FOR VALUE)

CIRCUIT DIAGRAM AND MARKING



INCHES	MM
.010	25
031	79
040	1 02
094	2 39
.187	4 75
281	7 14
.400	10.16
.600	15.24
.750	19.05
781	19.84
1.063	27.00

NOTES
1 Dimensions are in inches.
2 Metric equivalents are given for general information only and are based upon 1 00 inch = 25.4 mm.
3 Marking shall be on the side of the case opposite terminal pins.

FIGURE 1 Dimensions and configuration

PSC 5950

Source: https://assist.dla.mil -- Downloaded: 202610-16T03:44Z
Check the source to verify that this is the current version before use.

REQUIREMENTS (When numbers in parentheses, i e (1-2) are used, they indicate the winding and the extreme terminals of the winding)

Electrical ratings See table I.
Working voltage 350 volts peak

Design and construction
Dimensions and configuration See figure 1.
Case Plastic
Material: Diallyl phthalate
Terminals Pin, type D, per MIL-STD-1276, tinned.
Weight 1 2 ounce maximum.
Operating temperature range -55°C to +105°C
Altitude 75,000 feet maximum

Terminal strength MIL-STD-202, method 211, test condition A, 2 pounds

Dielectric withstanding voltage
At sea level 1,000 Vrms
At reduced barometric pressure 440 Vrms

Vibration MIL-SID-202, method 204, test condition D

Marking location See figure 1.

Part number M27/168- (dash number from table 1)

Source: https://assist.dla.mil -- Downloaded: 202610-16T03:44Z
Check the source to verify that this is the current version before use.

TABLE I Electrical ratings

Dash no	(1-2) 1/ Inductance at 1 volt		(1-2) 2/ DC current (max) (mA)	(1-2) DC resistance ±20% at 25°C (ohms)	(1-2) Quality factor		
	L ±2% (H)	Frequency (Hz)			Q min	Voltage (Vrms)	Frequency (Hz)
01	10	1,000	55	14	22	1	1,500
02	15	↑	45	22	↑	↑	↑
03	30		28	34			
04	50	↓	23	54	↓		↓
05	1 0	1,000	16	130	22		1,500
06	2 0	400	8	130	20		750
07	5 0	↑	5	340	↑		↑
08	7 5		4 5	517			
09	20		2 7	1,310	↓		
10	50	↓	1 4	3,180	20	↓	↓
11	100	400	1 1	8,550	21	1	750

1/ Inductance is measured at 0 DC
2/ The DC current is that amount that will reduce the inductance approximately 5% (8% maximum)

Custodians
Army - EL
Navy - EC
Air Force - 85

Review activities
Army - MI, SG
Navy - SH, OS
Air Force - 11, 17, 99
DLA - ES

User activities
Army - MU, WC, SG, ME
Navy - MC, AS
Air Force - 19

Preparing activity
Army - FI

Agent
DLA - ES

(Project 5950-0538-27)

Source: https://assist.dla.mil -- Downloaded: 202640-16T03:44Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION MIL-T-27/168 TRANSFORMERS AND INDUCTORS (AUDIO,POWER AND HIGH-POWER PULSE), INDUCTORS, HIGH Q, TF5R20ZZ

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

ESC-FM 1068-68

To detach this form, cut along this line

Source: https://assist.dla.mil -- Downloaded: 202640-16T03:44Z
Check the source to verify that this is the current version before use.

FOLD

Department of the Army
Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Commanding General
U.S.Army Electronics Command
ATTN: XXXXXXXX DRSEL-RD-TS
Fort Monmouth, New Jersey 07703

FOLD

To detach this form, cut along this line

Source: https://assist.dla.mil -- Downloaded: 202640-16T03:44Z
Check the source to verify that this is the current version before use.